O.C.E.A.N.

Ocean Contamination & Environmental Analysis Network

DIGITAL ENFORCEMENT ANALYSIS DOSSIER

Case ID	OCN-2026-0907-190458
Incident	oil_spill
Status	ANALYSIS COMPLETE
Generated	07-09-2026 19:04:58 India Standard Time
Document Reference	OCEAN_Enforcement_Dossier_2026-09-07_19-04-58.pdf

System-generated analytical dossier. Intended to support human review and investigation; it does not by itself constitute a legal determination or proof of liability.

1. EXECUTIVE CASE SUMMARY

Field	Value
Case ID	OCN-2026-0907-190458
Detection Time	2026-08-14T03:14:00Z
Analysis Time	07-09-2026 19:04:58 India Standard Time
Classification	oil_spill
Confidence	52.7%
Estimated Spill Area	105.96 km2
Oil Coverage	13.12%
Detected Pixels	8600
Detection Method	Enhanced SAR class-aware dark-patch
Image Source	class_1_00004.jpg
Estimated Origin	17.56228 N, 73.74121 E
Lead Suspect	M/T STENA NORDIC
VSI Score	87

Findings summary: the available artifacts record the classification and detection measurements above. The attribution section reflects the session data supplied by the active demonstration interface and is not independently verified AIS evidence.

2. AI-BASED SPILL DETECTION

Measure	Runtime value
Detection Classification	oil_spill
Confidence	52.7%
Oil Coverage	13.12%
Detected Pixels	8600
Estimated Area	105.96 km2
Detection Method	Enhanced SAR class-aware dark-patch

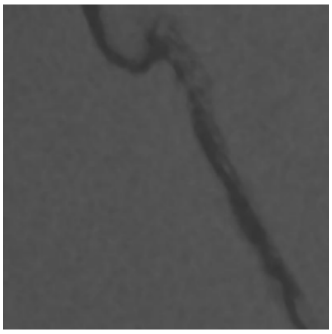

Figure 2.1 - Original satellite image (spill_original.png)

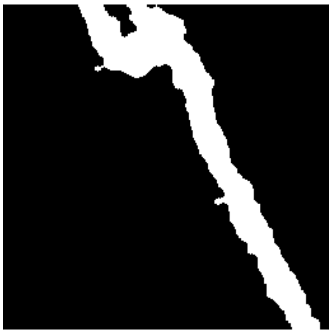

Figure 2.2 - Raw segmentation mask (spill_mask_raw.png)

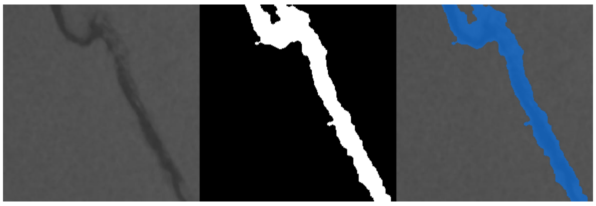

Figure 2.3 - Composite mask preview (spill_mask.png)

Figure 2.4 - Spill overlay (spill_overlay.png)

3. SPILL GEOMETRY & LOCATION

Field	Value
Geometry type	Polygon
Feature count	1
Bounding extent	Lon 73.23900 to 73.42900; Lat 17.16400 to 17.42000
Metadata	{"label": "Oil Spill Slick", "confidence": 52.7, "pixel_count": 8600, "detection_method": "Enhanced SAR class-aware dark-patch", "image_source": "class_1_00004.jpg", "detection_time_utc": "2026-08-14T03:14:00Z"}

Source artifact: static/spill_boundary.geojson. The source GeoJSON is read without modification.

4. HYDRODYNAMIC HINDCAST ANALYSIS

Parameter	Configured/runtime value
Backward window	36 hours
Monte Carlo particles	500
Snapshot interval	3 hours
Integration method	RK2 midpoint backward integration
Current dataset	physics_engine\data\cmems_currents.nc
Wind dataset	physics_engine\data\era5_winds.nc
Release-time uncertainty	+/- 30 minutes

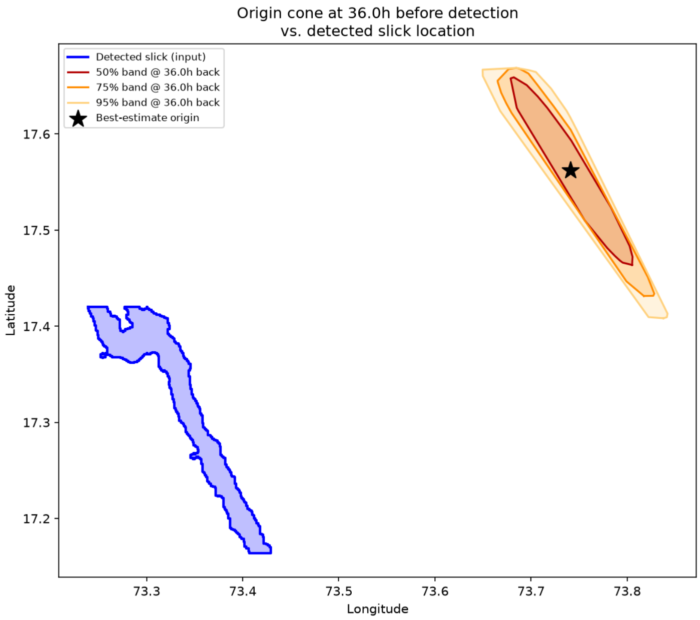

Figure 4.1 - Hindcast simulation verification (static/sanity_check_plot.png)

5. PROBABILISTIC SPILL ORIGIN

Measure	Value
Best-estimate origin	17.56228 N, 73.74121 E
Probability bands	50, 75, 95
Trajectory points	13
Detection time	2026-08-14T03:14:00Z
Source	static/origin_cone.geojson

The origin cone and uncertainty regions are reported from the existing hindcast GeoJSON. No origin coordinates are recalculated by the dossier generator.

6. AIS VESSEL ATTRIBUTION

Rank	Vessel	IMO	Type	Distance	Speed behavior	AIS status	VSI
1	M/T STENA NORDIC	9332237	Crude oil tanker	0.18 km	4.5 kn · AIS Off · Drifting Post-Discharge	AIS DARK (EVASION)	87
2	MV PACIFIC DAWN	9447210	Container ship	19.8 km	17.8 kn · Normal Transit	AIS ACTIVE	28
3	MV OCEAN TRADER	9558731	Bulk carrier	19.8 km	12.1 kn · AIS Active	AIS ACTIVE	34
4	MV COASTAL STAR	9601452	Product tanker	19.8 km	12.8 kn · Transit	AIS ACTIVE	41
5	FV HARBOUR LIGHT	9012345	Fishing vessel	19.8 km	7.2 kn · Trawling	AIS ACTIVE	12

Attribution source: current session data supplied by the existing demo interface. This dossier does not represent a live AIS archive query.

7. LEAD SUSPECT ANALYSIS

Field	Current session value
Vessel	M/T STENA NORDIC
IMO	9332237
Type	Crude oil tanker
VSI score	87

Factor	Observation	Weight	Contribution / score
F_loc — Distance to origin	0.18 km	40%	Not numerically available
F_spd — Speed anomaly	Slowed 7.2 → 1.4 kn	25%	Not numerically available
F_gap — AIS evasion	Dark 13:48–14:35Z	25%	Not numerically available
F_prof — Vessel profile	High risk (crude tanker)	10%	Not numerically available

8. EVIDENCE & ANALYTICAL BASIS

Category	Evidence item	Source	Analytical significance
Satellite / AI Detection	Classification and segmentation statistics	spill_analysis.json; mask artifacts	Supports the recorded spill classification
Spill Geometry	Detected boundary polygon	spill_boundary.geojson	Defines the analyzed slick geometry
Hydrodynamic Hindcast	Backward ensemble and plot	origin_cone.geojson; sanity_check_plot.png	Indicates a modeled origin region
AIS / Vessel Attribution	Current vessel session records	demo.html runtime state	Configured analytical indicators; not live verification
Behavioral Indicators	Speed and AIS observations	Current attribution payload	Supports prioritization for human review

9. ANALYTICAL FINDINGS

Detected incident: oil_spill.

Estimated origin: 17.56228 N, 73.74121 E.

Detection confidence: 52.7%.

Lead candidate: M/T STENA NORDIC; VSI 87.

The attribution values above are demo / configured attribution data and should be treated as analytical indicators pending qualified human review.

10. TECHNICAL METHODOLOGY

Image Input -> Spill Detection -> Segmentation / Classification -> Spill Boundary Extraction -> Hydrodynamic Hindcast -> Probabilistic Origin Estimation -> AIS Attribution -> Vessel Suspicion Assessment -> Enforcement Dossier

Implemented components used by this dossier include Flask, Python, the existing AI/classical detection pipeline, GeoJSON artifacts, the existing hindcast engine, Monte Carlo ensemble processing, RK2 integration, ReportLab, and Pillow image preparation.

11. DATA PROVENANCE, LIMITATIONS & HUMAN REVIEW

Sources used: static/spill_analysis.json, static/spill_boundary.geojson, static/origin_cone.geojson, available detection image artifacts, available hindcast plot, physics_engine configuration/data references, and the current frontend attribution payload.

Limitations: model, data, spatial, temporal, and attribution uncertainty may affect results. AIS attribution in the current interface is demo / configured data rather than a verified live archive query. Missing artifacts are omitted or reported as unavailable.

This dossier is system-generated from the analytical outputs available at the time of report generation. Results may contain model, data, spatial, temporal, or attribution uncertainty. The dossier is intended to support qualified human review and investigation and should not be interpreted as an independent legal finding, final attribution, or proof of liability.

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